

WESTERN SYDNEY
UNIVERSITY



Module 6

Subplots and Other Two-Dimensional Plots

Subplots

- Subplot command allows you to put multiple graphs on one figure window.
- `subplot(m,n,p)` divides figure window into a grid of m rows and n columns.
- Variable **p** identifies the part of the window where the plot is placed.

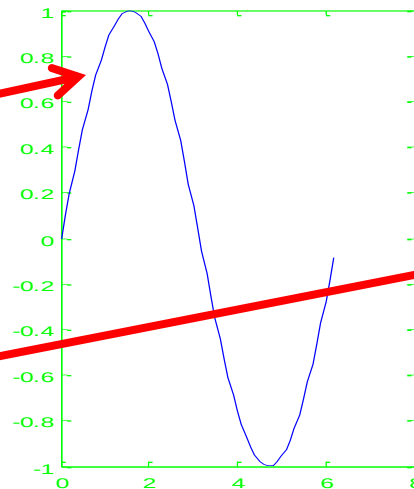
p = 1	p = 2
p = 3	p = 4

Examples of subplots

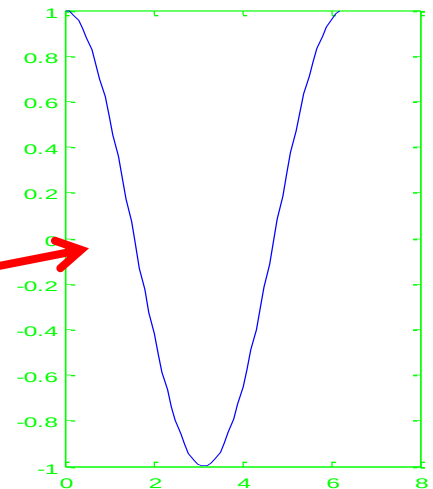
- To graph $\sin(x)$ and $\cos(x)$ on the same figure side-by-side do the following:

```
x = 0:0.1:2*pi;  
subplot(1,2,1);  
plot(x, sin(x));  
subplot(1,2,2);  
plot(x, cos(x));
```

this is position 1



this is position 2



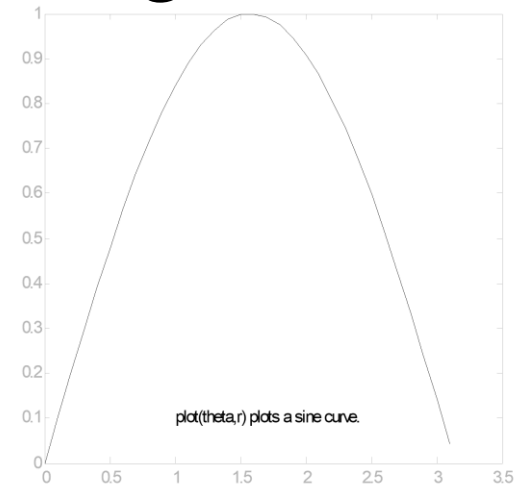
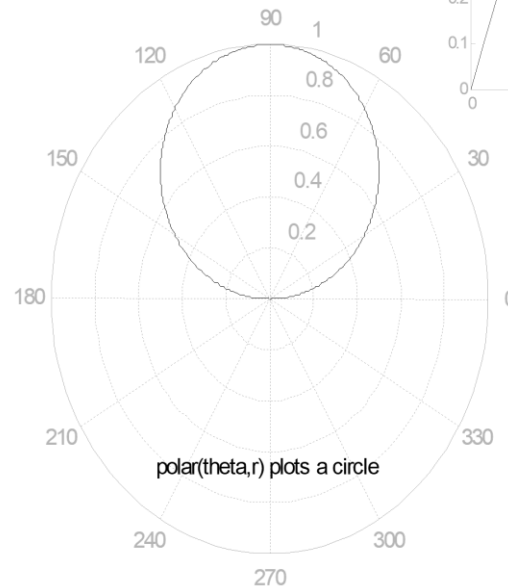
Other utilities for 2-D plots

- MATLAB supports several 2-D plotting utilities:
 - Polar plots
 - Logarithmic plots
 - Bar graphs
 - Pie charts

Polar plots

- MATLAB supports tools for plotting data in **polar coordinates**.

```
theta = linspace(0,pi,1000);  
r = sin(theta);  
plot(theta,r)  
polarplot(theta,r)
```

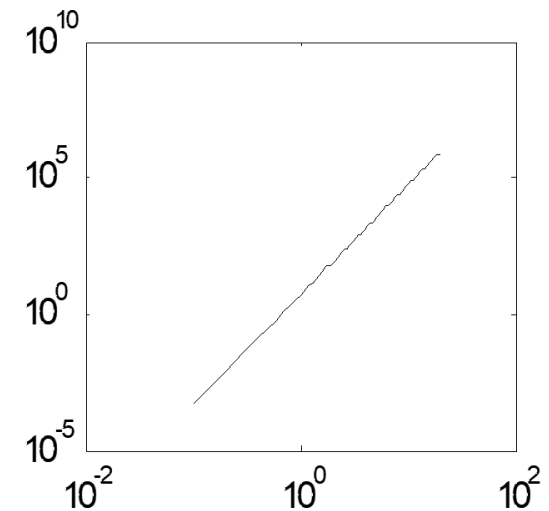
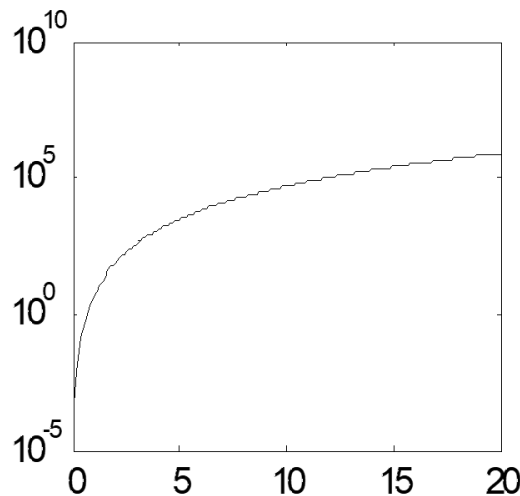
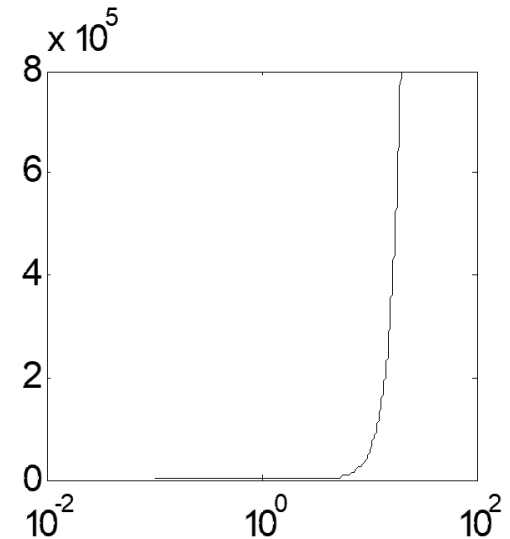
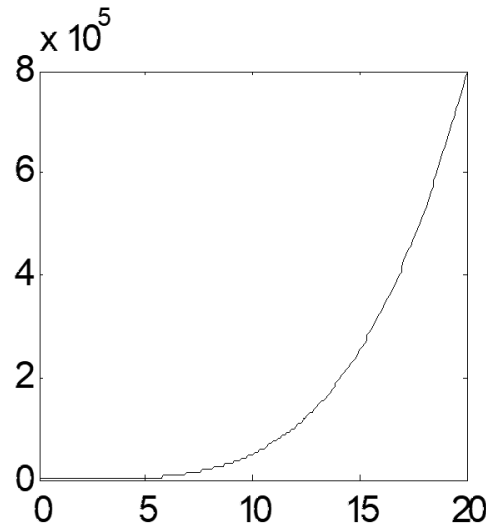


Logarithmic plots

- MATLAB has tools for three kinds of **logarithmic plots**:
 - **semilogx**
 - **semilogy**
 - **loglog**
- These plotting utilities automatically replace linear scales with logarithmic scales.
- Logarithmic scales are useful when a variable ranges over many orders of magnitude.

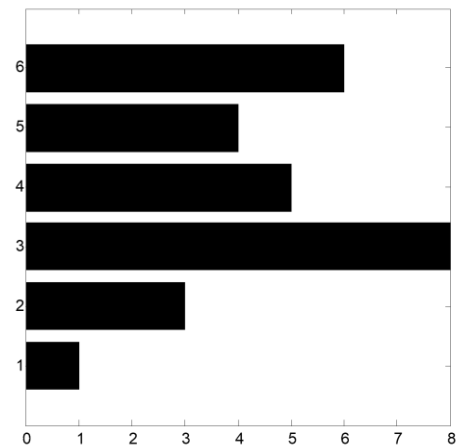
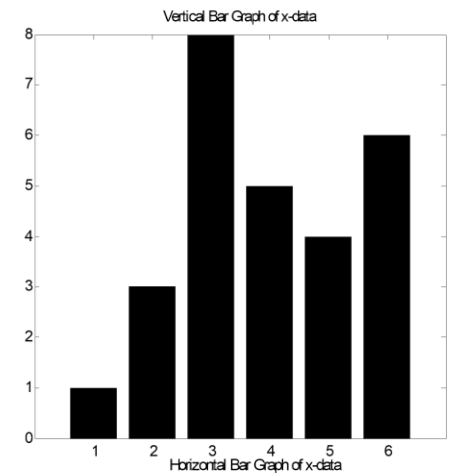
Example: Logarithmic plots & subplots

```
x = 0:0.1:20;  
y = 5*x.^4;  
subplot(2,2,1);  
plot(x,y);  
subplot(2,2,2);  
semilogx(x,y);  
subplot(2,2,3);  
semilogy(x,y);  
subplot(2,2,4);  
loglog(x,y);
```



Bar charts

- Bar graphs are useful for reporting data.
- $x = [1,3,8,5,4,6]$;
- `bar(x)` ; generates a vertical bar graph.
- `barh(x)` ; generates a horizontal bar graph.



Pie charts

- Pie charts are another useful way of reporting data.
- **pie(x)** ;
 $8/(1+3+8+5+4+6) \approx 30\%$
(cyan section)
 $5/(1+3+8+5+4+6) \approx 19\%$
(yellow section)
etc.

